Notes: Goldberg and Hellerstein (RES, 2013)

A Structural Approach to Identifying the Sources of Local Currency Price Stability

Contents

1 Big picture

- **Question.** Why do local-currency prices of imported consumer goods move so little after exchange-rate changes? The paper asks how much incomplete exchange-rate pass-through is due to local non-traded costs, markup adjustment, and nominal price rigidities.
- **Core puzzle.** In the beer market studied here, simple descriptive regressions imply that only around 5–10 percent of exchange-rate movements pass through to consumer prices, so roughly 90 percent of the exchange-rate shock is absorbed before it reaches retail prices.
- **Setting.** The authors study weekly store-level retail and wholesale prices for beer sold by Dominick's Finer Foods in the Chicago market from June 1991 to June 1995. The sample contains 16 beer brands and two price zones.
- **Empirical object.** The object is not just average pass-through. The paper decomposes the missing pass-through into: manufacturer local non-traded costs, manufacturer markup adjustment, indirect effects of competitors' repricing costs, own repricing costs, retailer local costs, retailer markup adjustment, and retailer repricing costs.
- **Main result.** On average, about 60 percent of incomplete pass-through is due to local non-traded costs, 8.2 percent to markup adjustment, 30.6 percent to own-brand price adjustment costs, and 1.2 percent to indirect or strategic effects of competitors' repricing costs.
- **Key economic intuition.** Exchange rates affect the traded component of costs, but many costs of selling imported goods in the destination market are local and dollar-denominated. Even when prices adjust, therefore, only part of marginal cost moves with exchange rates. In addition, prices sometimes do not adjust at all, which requires nominal rigidities to explain.
- **Why beer is useful.** Imported beers are directly exposed to exchange-rate shocks, weekly data reveal exact price changes and non-changes, and the data include both wholesale and retail prices, allowing the paper to separate manufacturer and retailer behavior.
- **Takeaway.** Local currency price stability is not a single mechanism. It is a combination of local-cost insulation and state-dependent nominal price adjustment, with the strongest nominal rigidity operating at the wholesale level.

2 Incremental contribution of the paper

- **From pass-through measurement to pass-through decomposition.** Earlier work documents incomplete exchange-rate pass-through. This paper decomposes the source of the incomplete response using a structural model of demand, vertical pricing, costs, markups, and price adjustment.
- **Adds nominal rigidities to structural pass-through accounting.** The previous micro pass-through literature could account for markup adjustment and local costs, but usually assumed static first-order conditions in every period. This paper explicitly allows firms not to satisfy static FOCs in periods when prices do not change.
- **Uses high-frequency data to identify non-adjustment.** Weekly data make it possible to observe long stretches of unchanged prices and discrete price jumps. Monthly or annual data

would mask this state-dependent adjustment margin.

- **Combines revealed preference with IO structure.** The paper derives upper and lower bounds for price adjustment costs: if a firm changes price, the gain from changing must exceed the adjustment cost; if it does not change price, the gain from changing must be smaller than the adjustment cost.
- **Separates wholesale and retail mechanisms.** Because both wholesale and retail prices are observed, the authors show that retail prices almost always move when wholesale prices move. The core rigidity therefore comes primarily from wholesale price setting, not retailer failure to pass through wholesale cost changes.
- **Quantifies direct and strategic rigidity effects.** Own-brand adjustment costs directly prevent price changes; competitors' adjustment costs also matter strategically because firms choose prices conditional on rivals' prices.
- **Shows local costs remain important even with menu costs.** A central lesson is that nominal price rigidities do not eliminate the role of local costs. Even after allowing for sticky prices, local non-traded costs remain the largest component of incomplete pass-through.
- **Provides a transferable method.** The specific demand and supply assumptions are tailored to beer, but the broader framework can be applied to other industries with high-frequency product-level prices and market shares.

3 Data and measurement

3.1 Market and data source

- **Retailer.** Dominick's Finer Foods, the second largest supermarket chain in Chicago in the mid-1990s.
- **Period.** Weekly observations from June 6, 1991 to June 1, 1995, giving 202 weeks.
- **Markets.** The authors aggregate stores into two price zones. A market is a price-zone-week.
- **Products.** A product is one six-pack serving of a brand of beer. The sample contains 16 brands and 6,464 product-market observations.
- **Prices.** The data include both retail prices and wholesale prices. This is crucial because it lets the authors identify whether incomplete pass-through happens before or after the retailer.
- **Supplementary variables.** The data are supplemented with manufacturer costs, product characteristics from Consumer Reports, advertising and cost information, exchange rates, wages, barley prices, electricity prices, and consumer demographics.

3.2 Main price facts

- Average retail price is about \$5.44 per six-pack, while average wholesale price is about \$4.36.
- Retail prices change in about 20 percent of product-week observations; wholesale prices change in only about 8 percent.

- Wholesale prices are much stickier than retail prices.
- When wholesale prices change, retail prices almost always change too. This is important because it suggests retail price rigidity is not the main source of local currency stability.
- Simple OLS price regressions find exchange-rate pass-through of about 6.7 percent to retail prices and 4.7 percent to wholesale prices.

3.3 Pass-through mechanisms

The paper studies three mechanisms:

1. **Local non-traded costs.** These are destination-market costs paid in dollars and not directly affected by exchange-rate shocks. They include local distribution, labor, and other local services.
2. **Markup adjustment.** Manufacturers or retailers may absorb exchange-rate changes by changing markups.
3. **Nominal price rigidities.** Fixed costs of changing prices may cause firms to keep prices unchanged even when static optimal prices move.

3.4 Measurement philosophy

- The paper treats adjustment costs broadly: they include not only literal menu costs but also managerial time, advertising and communication costs, customer-relations costs, and the option value of waiting.
- The empirical goal is not to identify the physical nature of the adjustment cost. The goal is to measure how much price inertia requires such a cost in a revealed-preference sense.
- Price adjustment costs alone are not the final object. The important object is how much incomplete pass-through they generate in counterfactual simulations.

4 Model and empirical specifications

4.1 Retailer problem

The retailer sells all differentiated products and chooses retail prices taking wholesale prices as given. Retail profits in market t are:

$$\Pi_t^r = \sum_j (p_{jt}^r - p_{jt}^w - ntc_{jt}^r) s_{jt}(p_t^r) - A_{jt}^r.$$

- p_{jt}^r is the retail price of product j.
- p_{jt}^w is the wholesale price paid by the retailer.
- ntc_{jt}^r is the retailer's local non-traded cost.
- $s_{jt}(p_t^r)$ is market share or quantity demanded.
- A_{jt}^r is the fixed cost of changing the retail price.

If the retail price changes, the retailer satisfies the static FOC:

$$s_{jt} + \sum_k (p^r_{kt} - p^w_{kt} - ntc^r_{kt}) \frac{\partial s_{kt}}{\partial p^r_{jt}} = 0.$$

In vector form, the pricing equation is:

$$p^r_t = p^w_t + ntc^r_t - \Omega^{-1}_{rt} s_t.$$

Interpretation. Retail price equals wholesale price plus local retail cost plus retail markup. If the price does not change, the FOC need not hold; instead, a revealed-preference inequality bounds the adjustment cost.

4.2 Manufacturer problem

Manufacturers are differentiated-product Bertrand oligopolists. A manufacturer w chooses wholesale prices for products $j \in S_w$:

$$\Pi^w_{jt} = \sum_{j \in S_w} (p^w_{jt} - c^w_{jt}) s_{jt}(p^r_t(p^w_t)) - A^w_{jt}.$$

The manufacturer marginal cost has traded and non-traded components:

$$c^w_{jt} = c^w_{jt}(tc^w_{jt}, ntc^w_{jt}).$$

- The traded component is paid in the producer's currency and moves with exchange rates.
- The non-traded component is paid in the destination currency and is insulated from exchange-rate shocks.
- If the wholesale price changes, the manufacturer satisfies a wholesale-price FOC.
- If the wholesale price does not change, a revealed-preference inequality bounds the manufacturer's price adjustment cost.

4.3 Demand system

Demand is estimated using a Berry-Levinsohn-Pakes random coefficients logit model. Consumer i's indirect utility from product j at time t is:

$$u_{ijt} = x_{jt}\beta_i - \alpha_i p_{jt} + \xi_{jt} + \epsilon_{ijt}.$$

The model allows consumer tastes for price and characteristics to vary with demographics:

$$\begin{pmatrix} \alpha_i \\ \beta_i \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \Pi D_i.$$

Market shares are simulated as:

$$s_{jt} = \frac{1}{N} \sum_{i=1}^{N} \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_k \exp(\delta_{kt} + \mu_{ikt})}.$$

Identification of demand. Prices are instrumented with cost shifters: domestic beverage wages interacted with brand fixed effects and weekly nominal exchange rates for foreign brands.

4.4 Cost decomposition

Manufacturer marginal costs are parameterized as:

$$\ln c^{w}_{jt} = \theta_j + \theta_{dw} \ln w^{d}_{t} + F_j \theta_{fw} \ln(e_{jt} w^{f}_{t}) + D_j \theta_{dp} \ln(pb^{d}_{t}) + F_j \theta_{fp} \ln(e_{jt} pb_{jt}) + \omega_{jt}.$$

- w^{d}_{t} is domestic U.S. wages.
- w^{f}_{t} is foreign wages.
- e_{jt} is the bilateral exchange rate.
- pb_{jt} is the barley price.
- The domestic-wage coefficient for foreign manufacturers identifies the local non-traded cost share.

5 Identification and empirical design

5.1 What identifies demand and markups

- Demand is identified from variation in relative prices and market shares across products, zones, and weeks.
- Exchange rates and input costs shift prices but are plausibly unrelated to unobserved time-varying product appeal.
- Once demand is estimated, the demand derivatives identify markups through the retailer and manufacturer first-order conditions.
- Marginal costs are recovered as price minus markup.

5.2 What identifies local non-traded costs

- If costs in dollars are stable even when exchange rates move, and if markups do not adjust enough to explain price stability, then a local non-traded cost component is needed.
- The cost function separates domestic U.S. costs from foreign-currency costs.
- The estimated domestic-cost share for foreign manufacturers is large, implying that much of the dollar marginal cost does not mechanically move with exchange rates.

5.3 What identifies price adjustment costs

- In periods with price changes, the firm must gain enough profit from changing price to cover the adjustment cost. This provides an upper bound.
- In periods without price changes, the firm must gain too little from changing price to justify paying the adjustment cost. This provides a lower bound.
- The key identifying variation is the high-frequency observation of price changes and non-changes at the product-week level.
- The estimates are bounds, not exact costs. The counterfactual simulations use upper bounds to give nominal rigidities the largest plausible role.

5.4 Interpretation of causal claims

- The paper is structural rather than reduced-form causal in the experimental sense.
- The credibility comes from the combination of a flexible demand model, observed wholesale and retail prices, cost shifters, and revealed-preference inequalities.
- The decomposition depends on the model of demand, vertical pricing, and static myopic firm behavior.
- The authors address dynamic concerns by arguing that beer inventories are limited and exchange-rate shocks are persistent and difficult to forecast.

6 Main empirical results

6.1 Descriptive pass-through

- Retail prices respond only weakly to exchange rates: the estimated retail pass-through elasticity is about 6.7 percent.
- Wholesale price pass-through is even lower, about 4.7 percent.
- The retail price responds almost fully to wholesale price movements, suggesting the retail stage does not introduce the main block to pass-through.

6.2 Demand estimation

- The simple multinomial logit diagnostic shows that IV estimates make consumers more price-sensitive than OLS estimates.
- The random coefficients model allows income to interact with price sensitivity and alcohol content.
- Higher-income consumers are less price sensitive, and demand curvature is flexible enough to generate markup adjustment.
- The estimated super-elasticity is around 0.8, meaning a 10 percent price increase raises the absolute elasticity of demand by about 8 percent.

6.3 Costs and markups

- Retail and manufacturer markups are economically reasonable, around 5–10 percent of retail price.
- The retailer's non-traded costs are around 30–40 cents per six-pack for most foreign brands.
- Manufacturer local non-traded costs are large: the domestic U.S. wage share estimate is 0.61.
- This means that even imported beer has a substantial U.S.-denominated cost component.

6.4 Adjustment costs

- Retail adjustment-cost bounds are meaningful but do not explain much pass-through failure, because retail prices typically change whenever wholesale prices change.
- Manufacturer adjustment-cost bounds are central because wholesale prices often remain fixed despite exchange-rate variation.
- Permanent level price changes have much higher retail adjustment costs than temporary sales.

6.5 Counterfactual decomposition

- With no repricing costs, median retail pass-through is around 26.2 percent, far above the 5–10 percent observed in the data.
- If competitors' prices remain fixed due to repricing costs, median retail pass-through falls to about 18 percent.
- If own repricing costs are included, median manufacturer and retail pass-through can fall to zero in the median counterfactual.
- The final decomposition assigns 60 percent of incomplete pass-through to local non-traded costs, 8.2 percent to markup adjustment, 30.6 percent to own repricing costs, and 1.2 percent to strategic repricing-cost effects.

7 Tables and figures

7.1 Table 1: Summary statistics for prices and product characteristics

Read:

- Table 1 reports the core price and product-characteristic variables used in the empirical analysis.
- The mean retail price is $5.44 per six-pack, while the mean wholesale price is $4.36.
- Retail prices change in 20 percent of observations, but wholesale prices change in only 8 percent.
- The price-change frequencies show that price rigidity is present at both stages, but is stronger at the wholesale stage.
- Product characteristics include percent alcohol and bitterness, which enter the demand system.

Economic message: The table establishes the empirical setting: high-frequency data show many price non-adjustment periods, and the availability of both retail and wholesale prices allows the paper to locate where price rigidity arises.

7.2 Table 2: Preliminary descriptive results

Read:

- Table 2 estimates simple price regressions of retail and wholesale prices on exchange rates and controls.
- The retail pass-through estimate is about 6–7 percent.

- The wholesale pass-through estimate is about 4–5 percent.
- The retail price responds almost one-for-one to wholesale prices.
- This implies that the missing pass-through is largely explained before the retail price reacts to the wholesale price.

Economic message: The descriptive evidence motivates the structural model: pass-through is very low, and the main source of inertia appears to be upstream, at the wholesale/manufacturer stage.

TABLE 1
Summary statistics for prices and product characteristics for the 16 products in the sample

Category	Description	Mean	Median	Standard deviation
Prices	Retail prices ($ per six-pack)	5·44	5·79	1·28
	Wholesale prices ($ per six-pack)	4·36	4·61	1·00
	Dummy for retail price change (=1 if yes)	0·20	0	0·40
	Dummy for wholesale price change (=1 if yes)	0·08	0	0·27
Product characteristics	Percent alcohol	4·52	4·60	0·68
	Bitterness	2·50	2·10	1·08

Notes: 6464 observations. *Source:* Dominick's; "Beer Ratings." Consumer Reports (1996, pp. 10–19).

(a) (a) Table 1: Summary statistics

TABLE 2
Some preliminary descriptive results

	Retail price	Retail price	Wholesale price	Wholesale price	Retail price
Exchange rate	5·96 (1·50)**	6·72 (1·56)**	4·27 (1·50)**	4·74 (1·52)**	
Wholesale price					105·37 (2·53)**
R^2	0·65	0·65	0·81	0·81	0·80

Notes: The dependent variable is the retail or the wholesale price for a six-pack of each brand of beer. The exchange rate is the average of the previous week's bilateral spot rate between the foreign manufacturer's country and the U.S. (dollars

(b) (b) Table 2: Preliminary price regressions

7.3 Figures 1 and 2: Weekly prices for Bass Ale and Corona

Read:

- Figure 1 plots weekly retail and wholesale prices for Bass Ale.
- Figure 2 plots weekly retail and wholesale prices for Corona.
- The figures show long flat segments followed by discrete jumps, the classic empirical pattern associated with state-dependent price adjustment.
- The graphs also show temporary price reductions or sales, followed by returns to the prior regular price.
- Retail prices move closely with wholesale prices; wholesale prices are the more persistent object.

Economic message: The figures visually justify the paper's focus on nominal rigidities. Prices are not smoothly adjusted every week; they stay fixed for stretches and then jump.

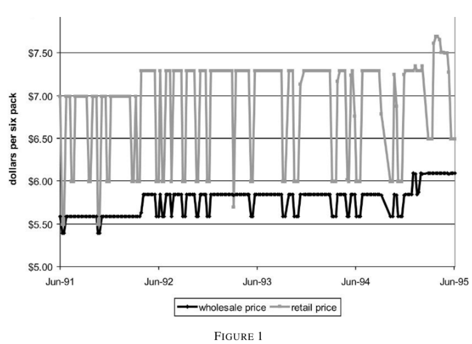

FIGURE 1
Weekly retail and wholesale prices for Bass Ale.
Prices are for a single six-pack and are from Zone 1. 202 observations. Source: *Dominick's*

price adjustment may generate significant price inertia, even if the nominal price rigidities facing each individual manufacturer or retailer are small.

(a) (a) Figure 1: Bass Ale weekly prices

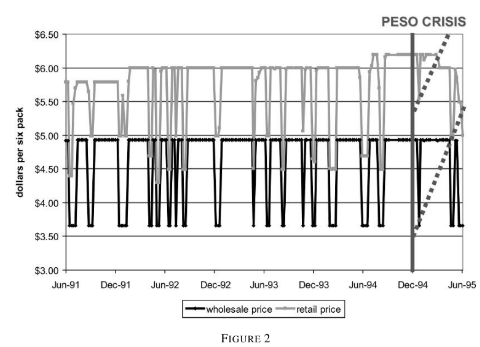

FIGURE 2
Weekly retail and wholesale prices for Corona.
Prices are for a single six-pack and are from Zone 1. 202 observations. Source: *Dominick's*

.4. *Discussion of the main assumptions*

(b) (b) Figure 2: Corona weekly prices

7.4 Table 3: Diagnostic multinomial logit demand estimates

Read:

- Table 3 uses a simpler multinomial logit model as a diagnostic before estimating the full random coefficients model.
- The OLS price coefficient is around -0.93, while the IV coefficient is around -2.43.
- Instrumenting for price makes consumers appear more price sensitive, consistent with upward bias in OLS due to unobserved product appeal.
- The first-stage partial F-statistics are above 34, suggesting the instruments are strong.

Economic message: The demand instruments are informative and move the price coefficient in the expected direction. This supports the use of cost and exchange-rate shifters in the full model.

7.5 Table 4: Full random coefficients logit demand estimates

Read:

- Table 4 reports the flexible demand system used in the structural analysis.
- The mean price coefficient is negative and significant.
- The interaction of price with income is positive, implying higher-income consumers are less price-sensitive.
- Bitterness and percent alcohol enter positively in mean utility.
- The income interaction with percent alcohol is negative, consistent with different tastes across income groups.

Economic message: Flexible demand estimation is essential because pass-through depends on demand curvature and markups. The model allows substitution patterns and price sensitivities to vary across consumers.

TABLE 3
Diagnostic results from the multinomial logit model of demand

Variable	OLS	OLS	IV	IV
Price	−0·93 (0·01)**	−0·92 (0·01)**	−2·43 (0·35)**	−2·43 (0·35)**
Holiday		0·06 (0·02)**		0·001 (0·01)
First-stage partial F-statistic			34·45	34·24

Notes: The dependent variable is $\ln(s_{jt}) - \ln(s_{0t})$. Regressions include brand fixed effects. Huber–White robust stan-

(a) (a) Table 3: Diagnostic logit demand

TABLE 4
Results from the full random coefficients logit model of demand

Variable	Mean in population	Interaction with income
Price	−2·48 (0·13)*	38·45 (12·20)*
Bitterness	0·81 (0·13)*	
Percent alcohol	0·74 (0·01)*	−50·36 (9·14)*
Minimum distance weighted R^2	0·84	

(b) (b) Table 4: Random coefficients demand

7.6 Table 5: Mean prices, markups, and costs for selected foreign brands

Read:

- Table 5 reports average retail and manufacturer prices, markups, and cost components for Bass, Beck's, Corona, Heineken, and all imports.
- Retail markups are about 40 cents per six-pack.
- Manufacturer markups are also economically modest, generally around 40–50 cents.
- Retail non-traded costs vary across brands but are generally around 30–50 cents.

- Manufacturer total costs account for most of the wholesale price.

Economic message: The markup estimates are reasonable and consistent with industry evidence. This helps validate the structural demand-and-supply model used to decompose pass-through.

associated with a price change that are available for the empirical analysis. Still, in the remainder of the paper, we report results based on this second more conservative approach.

Table 5 reports retail and wholesale prices and our estimates of markups and costs for selected foreign brands. The markups appear reasonable and consistent with industry wisdom, on

TABLE 5

Mean prices, markups, and costs for selected foreign brands

	Bass	Becks	Corona	Heineken	All imports
Retailer					
Price	6·36 (0·64)	5·27 (0·48)	5·10 (0·62)	5·66 (0·70)	5·52 (0·90)
Markup	0·41 (0·005)**	0·41 (0·003)**	0·40 (0·004)**	0·40 (0·004)**	0·41 (0·003)**
Non-traded costs					
Backed out	0·26 (0·070)**	0·41 (0·045)**	0·39 (0·091)**	0·47 (0·070)**	0·42 (0·038)**
Fitted	0·24 (0·071)**	0·38 (0·047)**	0·36 (0·091)**	0·45 (0·070)**	0·38 (0·041)**
Manufacturer					
Price	5·76 (0·24)	4·44 (0·16)	4·27 (0·43)	4·99 (0·28)	4·69 (0·77)
Markup	0·51 (0·005)**	0·40 (0·003)**	0·40 (0·003)**	0·53 (0·006)**	0·44 (0·003)**
Total costs					
Backed out	5·28 (0·04)**	4·04 (0·01)**	3·83 (0·07)**	4·59 (0·07)**	4·41 (0·05)**
Fitted	5·21 (0·03)**	4·04 (0·01)**	3·82 (0·05)**	4·48 (0·07)**	4·25 (0·02)**

Notes: In the upper panel of the table, each entry reports the mean across weeks and zones of the retailer's prices, derived markups, and derived backed-out or fitted non-traded costs by brand in dollars per six-pack. In the lower panel of the table, each entry reports the mean across weeks and zones of the manufacturer's prices, derived markups, and derived backed-out or fitted total costs by brand in dollars per six-pack. The markups are price less marginal cost with the marginal costs derived from the structural model. The numbers in parentheses under the prices are standard deviations over the sample, and under the other variables are standard errors from bootstrap simulations with 400 draws. Those starred are significant at the *5% or **1% level. *Source:* Dominick's; Authors' calculations.

Table 5: Prices, markups, and costs

7.7 Table 6: Retail non-traded cost regression

Read:

- Table 6 regresses backed-out retail non-traded costs on Chicago-area grocery wages.
- The wage coefficient is positive and highly significant.
- The regression is used to predict retail non-traded costs in weeks when retail prices do not change and retail FOCs cannot be used directly.

Economic message: Retail local costs are modeled as local wage-driven costs. This lets the authors fill in non-traded costs in non-adjustment periods.

7.8 Table 7: Bounds for retailer adjustment costs

Read:

- Table 7 reports upper and lower bounds on retailer adjustment costs for selected imported brands.
- Bounds are reported both in dollars and as a share of brand revenue.
- Across all imports, the upper-bound adjustment cost is about 3.13 percent of revenue; the lower bound is about 0.57 percent.

- These magnitudes are comparable to other estimates of repricing costs, although the paper's concept is broader than literal menu costs.

Economic message: Retail repricing costs exist, but because retail prices usually move when wholesale prices move, they are not the main source of incomplete pass-through.

TABLE 6

Results from regressions of backed-out retailer non-traded costs on determinants

Variable	Coefficient
Chicago area grocery wages	0·51 (0·05)**
R^2	0·14
Observations	805

Notes: Dependent variable is the retailer's non-traded cost, which varies by week. Huber–White robust standard errors are reported in parentheses. Those starred are

(a) (a) Table 6: Retail non-traded costs

TABLE 7

Bounds for the retailer's adjustment costs for selected foreign brands

	Mean cost		Share of brand's revenue	
Brand	Upper bounds ($)	Lower bounds ($)	Upper bounds (%)	Lower bounds (%)
Bass	$103·32 (58·45)*	$6·16 (2·40)**	3·05 (2·01)	0·18 (0·07)**
Beck's	$456·48 (273·89)*	$143·06 (9·21)**	4·25 (1·30)**	1·33 (0·06)**
Corona	$131·36 (70·95)*	$61·00 (31·99)*	1·09 (0·54)**	0·51 (0·20)*
Heineken	$621·12 (249·73)**	$142·28 (30·00)**	3·50 (1·77)**	0·80 (0·16)**
All	$249·26 (78·96)**	$45·36 (7·97)**	3·13 (1·08)**	0·57 (0·12)**

Notes: The entries in the first two columns report the mean over time of the dollar value of adjustment costs, and in the third and fourth columns, the mean over revenue for that brand over all markets whether the price changed or not. Standard errors from bootstrap simulations with 400 draws reported in parentheses under each coefficient. Starred coefficients are significant at the *10% or **5% level.

(b) (b) Table 7: Retail adjustment-cost bounds

7.9 Table 8: Manufacturer cost-function regression

Read:

- Table 8 estimates the manufacturer marginal-cost function using periods when wholesale prices change.
- The domestic U.S. wage coefficient is 0.61 and significant.
- Foreign barley and foreign wages also matter, with coefficients 0.17 and 0.22.
- The local U.S. cost share is therefore large for imported beer sold in the U.S.

Economic message: Imported beer is not purely a foreign-cost good once it is sold in the U.S. market. A large local dollar-denominated component mechanically limits exchange-rate pass-through.

7.10 Table 9: Bounds for manufacturer adjustment costs

Read:

- Table 9 reports adjustment-cost bounds for wholesale price changes.
- Across all imported brands, the upper bound is about $178.60 and the lower bound is about $85.32.
- As a share of revenue, the upper bound is about 1.28 percent and the lower bound about 0.61 percent.
- Manufacturer adjustment costs matter because wholesale prices often remain unchanged.

Economic message: Wholesale repricing costs are a central source of complete non-adjustment. Even if the costs are not enormous relative to revenue, they can prevent adjustment when profit gains from adjustment are small.

TABLE 8

Results from constrained linear regression of foreign manufacturer total backed-out costs on determinants

Variable	Coefficient
Domestic U.S. wages	0·61 (0·11)**
Price foreign barley	0·17 (0·05)**
Foreign wages	0·22 (0·06)**

(a) (a) Table 8: Manufacturer cost regression

TABLE 9

Bounds for manufacturers' adjustment costs for selected foreign brands

	Mean cost		Share of brand's revenue	
Brand	Upper bounds ($)	Lower bounds ($)	Upper bounds (%)	Lower bounds (%)
Bass	$36·76 (18·54)*	$12·86 (0·35)**	0·80 (0·39)*	0·28 (0·02)**
Beck's	$310·44 (121·45)**	$109·14 (21·25)**	2·43 (0·85)**	0·85 (0·15)**
Corona	$213·48 (35·55)**	$181·00 (7·61)**	1·41 (0·23)**	1·20 (0·01)**
Heineken	$122·60 (73·02)**	$112·38 (4·83)**	0·22 (0·10)**	0·20 (0·02)**
All	$178·60 (50·31)**	$85·32 (3·03)**	1·28 (0·34)**	0·61 (0·02)**

Notes: The entries in the first two columns report the mean over time of the dollar value of adjustment costs, and in the third and fourth columns, the mean over revenue for that brand over all markets whether the price changed or not. Standard errors from bootstrap simulations with 400 draws reported in parentheses under each coefficient. Starred coefficients are significant at the *10% or **5% level.

(b) (b) Table 9: Manufacturer adjustment-cost bounds

7.11 Table 10: Adjustment costs for level changes versus sales

Read:

- Table 10 compares adjustment costs associated with level retail-price changes and temporary sales.
- Level changes have much higher adjustment costs, around $852.75.
- Temporary price changes have much lower costs, around $50.20.
- This distinction helps explain why regular prices are sticky even when temporary promotions occur.

Economic message: Not all price changes are the same. Changing the regular price is much more costly than running a sale, consistent with the idea that regular-price changes require more managerial and coordination effort.

o assess how firms react to exchange rate shocks. We consider the effect c
:urrency appreciation on foreign brands' prices in three scenarios, each with a
nption about the nature of the repricing costs faced by foreign brands. We first c

TABLE 10

Regression of retailer's fixed adjustment costs on a dummy for a level retail-price change

Variable	Coefficient

Table 10: Level-change adjustment costs

7.12 Table 11: Counterfactual pass-through experiments

Read:

- Table 11 reports median pass-through after a 5 percent appreciation of the relevant foreign currency.
- With no repricing costs, median retail pass-through across all imports is 26.2 percent.
- If competitor brand repricing costs keep competitors' prices fixed, median retail pass-through falls to 18.0 percent.

- If own-brand repricing costs are included, median pass-through becomes zero in the reported counterfactual.
- The table separates manufacturer and retailer channels and shows that wholesale price adjustment is decisive.

Economic message: The simulations translate the model estimates into pass-through magnitudes. They show that local costs and wholesale repricing costs together can generate the low pass-through observed in the data.

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TABLE 11

Counterfactual experiments: median pass-through of a 5% appreciation of the relevant foreign currency

	Manufacturer		Retailer	
	Traded	Markup adjustment	Traded	Markup adjustment
No repricing costs				
Bass	40·0 (0·0)**	30·4 (7·4)**	26·3 (6·5)**	24·5 (7·9)**
Beck's	40·0 (0·0)**	38·3 (3·2)**	34·2 (3·7)**	29·3 (4·2)**
Corona	40·0 (0·0)**	31·9 (11·8)**	28·1 (7·0)**	25·7 (8·1)**
Heineken	40·0 (0·0)**	37·1 (5·6)**	31·1 (6·1)**	28·3 (5·9)**
All	40·0 (0·0)**	31·8 (2·9)**	28·4 (3·6)**	26·2 (4·3)**
Competitor brand repricing costs				
Bass	40·0 (0·0)**	0·0 (0·0)	0·0 (0·0)	0·0 (7·9)**
Becks	40·0 (0·0)**	35·8 (10·0)**	33·3 (7·3)**	18·0 (8·7)**
Corona	40·0 (0·0)**	31·5 (8·0)**	27·8 (8·9)**	18·0 (7·6)**
Heineken	40·0 (0·0)**	31·0 (6·7)**	25·6 (6·2)**	7·5 (2·3)**
All	40·0 (0·0)**	30·6 (3·3)**	25·7 (3·9)**	18·0 (4·1)**
Own brand repricing costs				
Bass	40·0 (0·0)**	0·0 (0·0)	0·0 (0·0)	0·0 (0·0)
Becks	40·0 (0·0)**	0·0 (0·0)	0·0 (0·1)	0·0 (0·0)
Corona	40·0 (0·0)**	0·0 (0·0)	0·0 (0·1)	0·0 (0·0)
Heineken	40·0 (0·0)**	0·0 (0·0)	0·0 (0·0)	0·0 (0·0)
All	40·0 (0·0)**	0·0 (0·0)	0·0 (0.0)	0·0 (0.0)

Notes: Median over 404 markets. Retailer's incomplete pass-through: the retail price's percent change for the given percent change in the exchange rate, attributed to the presence of local dollar-denominated costs or to the retailer's markup adjustment. Manufacturer's incomplete pass-through: the manufacturer price's percent change for a given percent change in the exchange rate, attributed to the share of local dollar-denominated costs in the manufacturer's total costs or to the manufacturer's markup adjustment. Standard errors from bootstrap simulations with 400 draws reported in parentheses under each coefficient. Starred coefficients significant at the *10% or **5% level. *Source:* Authors' calculations.

the median pass-through falls to 28·4% and ranges from 26·3% for *Bass* to 34·2% for *Beck's*. Finally, the median retailer pass-through elasticity across all brands is 26·2%. This counterfactual reveals that the curvature of demand is such that the retailer passes through the bulk of its cost shocks to its prices, rather than adjusting its markups. The median pass-through of the original shock to the retail price ranges from 24·5% for *Bass* to 29·3% for *Beck's*. Note that these results are larger than the 5–10% retail pass-through elasticities estimated in Section 2.2.

We turn next to the case where nominal price rigidities are present. Because firms in our

Table 11: Counterfactual pass-through experiments

7.13 Table 12: Decomposition of incomplete transmission

Read:

- Table 12 decomposes the incomplete transmission of a 5 percent foreign-currency appreciation.
- Across all imports, local manufacturer costs explain 60.0 percent of incomplete pass-through.
- Manufacturer markup adjustment explains 8.2 percent.
- Other brands' repricing costs explain 1.2 percent.

- Own-brand repricing costs explain 30.6 percent.
- Retail local costs, retail markup adjustment, and retail repricing costs explain essentially zero in the final decomposition.

Economic message: The table is the paper's main quantitative conclusion. Local non-traded costs are the largest source of incomplete pass-through, but nominal rigidity is also economically important, mainly through wholesale price adjustment.

TABLE 12

Counterfactual experiments: decomposition of the incomplete transmission of a 5% appreciation of the relevant foreign currency to consumer prices

	Manufacturer				Retailer				
	Local costs	Markup adjustment	Costs of repricing		Local costs	Markup adjustment	Costs of repricing		
Brand			Other	Own			Other	Own	Total
Bass	60·0	9·6	30·4	0·0	0·0	0·0	0·0	0·0	100·0
Beck's	60·0	1·7	2·5	35·8	0·0	0·0	0·0	0·0	100·0
Corona	60·0	8·1	0·4	31·5	0·0	0·0	0·0	0·0	100·0
Heineken	60·0	2·9	6·1	31·0	0·0	0·0	0·0	0·0	100·0
All	60·0	8·2	1·2	30·6	0·0	0·0	0·0	0·0	100·0

Notes: Median over 404 markets. Local costs: the share of the incomplete transmission explained by the presence of a local dollar-denominated component in foreign manufacturers' or the retailer's marginal costs. Markup adjustment: the share of the incomplete transmission explained by the retailer or manufacturer's markup adjustment excluding markup adjustment due to fixed costs of price adjustment. Repricing costs, other: The effect of competitors' costs of price adjustment on the manufacturer's or retailer's own price adjustment behaviour. Repricing costs own: fixed costs of price adjustment incurred by the manufacturer or retailer to change its own price. *Source:* Authors' calculations.

Table 12: Decomposition of incomplete transmission

8 Short summary

- This paper studies why local-currency prices of imported goods are stable after exchange-rate shocks.
- It uses weekly beer-market data from Dominick's with both wholesale and retail prices.
- The paper estimates a random coefficients demand system, backs out markups and costs using manufacturer and retailer FOCs, and uses revealed-preference inequalities to bound price adjustment costs.
- Simple regressions show very low exchange-rate pass-through: about 6.7 percent to retail prices and 4.7 percent to wholesale prices.
- Retail prices almost always adjust when wholesale prices adjust, so the primary inertia is upstream at the wholesale level.
- The estimated local-cost share in manufacturer costs is large, with domestic U.S. wages accounting for about 61 percent of foreign manufacturers' costs in the cost regression.
- Counterfactual simulations show that local non-traded costs explain about 60 percent of incomplete pass-through.
- Own-brand repricing costs explain about 30.6 percent, while markup adjustment explains about 8.2 percent.
- The paper's main contribution is to integrate nominal price rigidity into structural pass-through accounting and show that both local costs and wholesale repricing costs are needed to explain local currency price stability.

9 Conclusion

9.1 Core conclusion

Goldberg and Hellerstein show that local currency price stability cannot be understood only through markup adjustment or only through menu costs. In the beer market, most incomplete pass-through is due to local non-traded costs, but the fact that prices sometimes do not move at all requires nominal price rigidities, especially at the wholesale level.

9.2 Economic intuition

A foreign-currency appreciation raises the traded component of foreign manufacturers' costs. But the price of an imported consumer good also reflects local distribution, labor, retailing, and other destination-market costs. These local costs do not move with exchange rates. Therefore, even if firms reoptimize prices, pass-through is mechanically incomplete. On top of that, firms may choose not to change wholesale prices when the gain from changing is smaller than the adjustment cost.

9.3 Incremental contribution revisited

The paper's incremental contribution is methodological and substantive. Methodologically, it adds revealed-preference bounds on price adjustment costs to a structural pass-through model. Substantively, it shows that the previous emphasis on local costs is robust even after allowing for nominal rigidities, but also that nominal rigidities are too important to ignore.

9.4 Implications

- For international macroeconomics, the results imply that exchange-rate movements may have muted consumer-price effects because of local distribution costs and sticky wholesale prices.
- For industrial organization, the paper shows how high-frequency price data can identify price-adjustment frictions in a structural model.
- For monetary and exchange-rate policy, the findings suggest that devaluation or appreciation may affect consumer prices slowly and incompletely, especially for goods with large local cost components.

9.5 Limitations

- The model is static and myopic, although the authors provide arguments and robustness checks suggesting this is reasonable for exchange-rate shocks.
- The results come from one product category and one retail chain, so the magnitudes may not generalize mechanically to all imported goods.
- The adjustment-cost estimates are bounds and capture a broad residual concept of repricing frictions rather than literal menu costs alone.
- The vertical model abstracts from some institutional details of distributors, although the authors argue that resale price maintenance makes the simplification reasonable.

9.6 Takeaway

The paper's central lesson is that local currency price stability is generated by both real and nominal frictions. Local non-traded costs dampen pass-through even when prices adjust, while wholesale price adjustment costs explain why prices often do not adjust at all.